Assignment No. 14
Discharge coefficient: planner vs axisymmetric

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Objective

- Comparing co efficient of discharge for an orifice of planner and axisymmetry

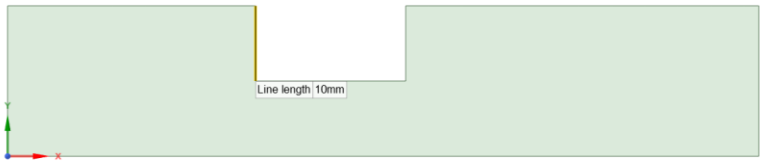
Problem Statement

- Coefficient of discharge $C_d = \frac{\dot{m}_{actual}}{\dot{m}_{theory}}$
- The theoretical mass flow rate for incompressible flow, $\dot{m}_{theory}$, can be obtained from Bernoulli's equation.

$$\dot{m}_{theory} = A_t\sqrt{2\rho(P_{upstream} - P_{downstream})}$$

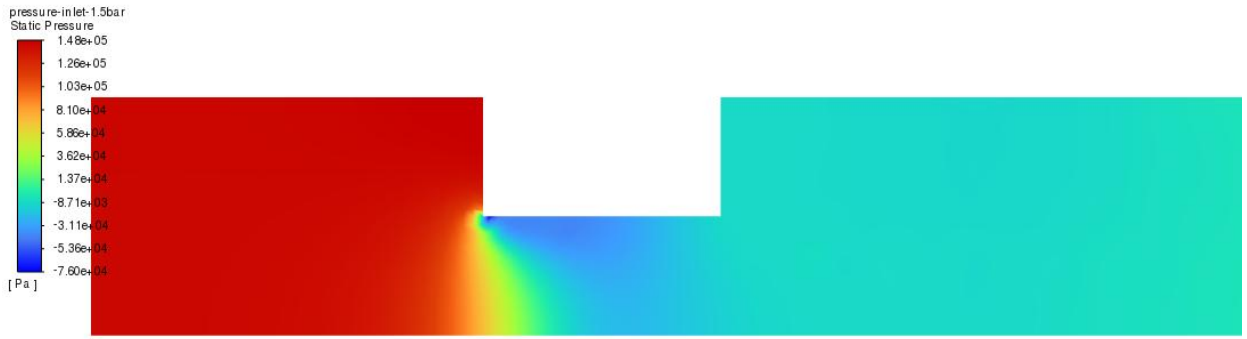
Axisymmetric Case

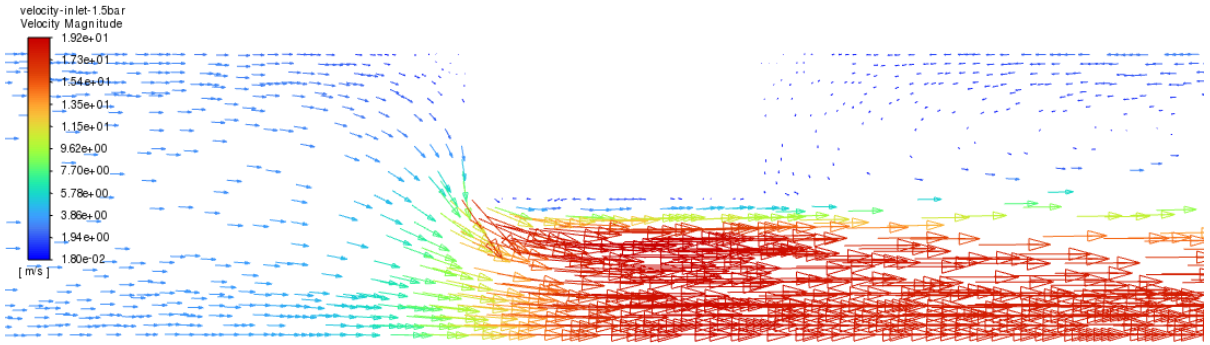
- The meshing ensured that there is atleast 10 cells at the throat region
- Using the default under relaxation factor resulted in very stable convergence



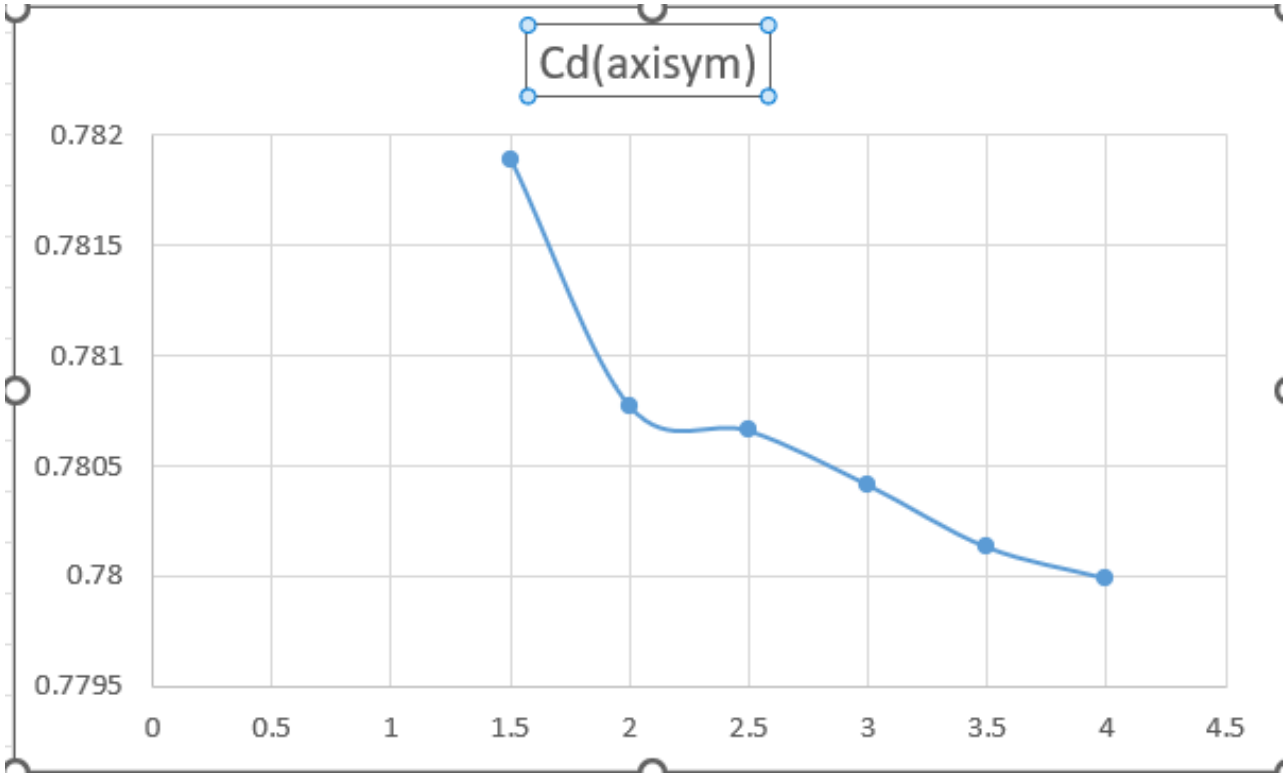
Results

For 1.5 bar total inlet pressure, pressure contour and velocity vector:



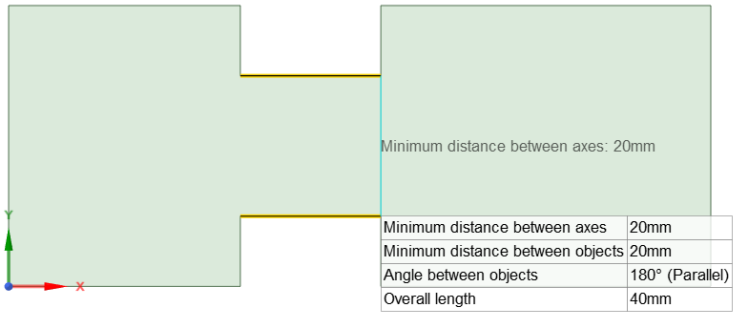


Co-efficient of discharge



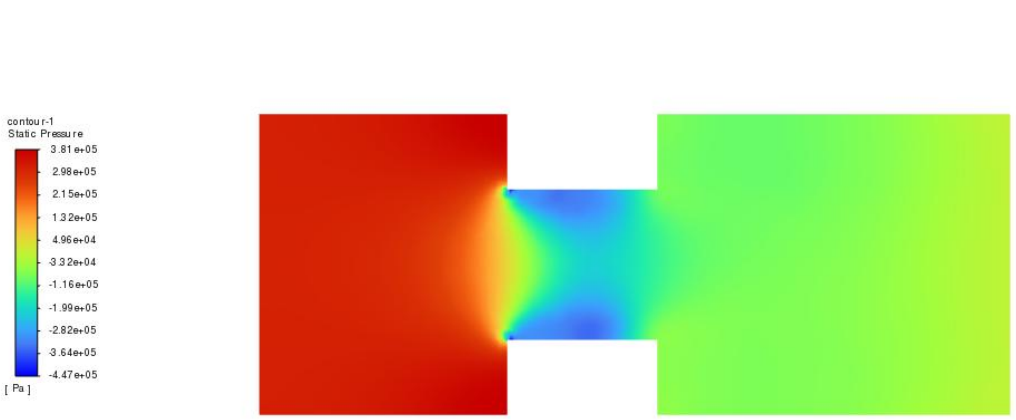
Planner Case

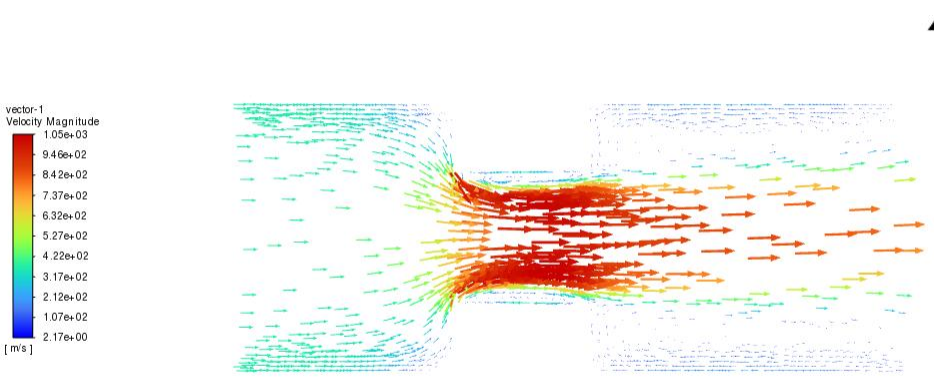
- The meshing ensured that there is atleast 20 cells at the throat region
- With Under relaxation factors, the solution was hardly converging. The solution somewhat converged with default pseudo time method values



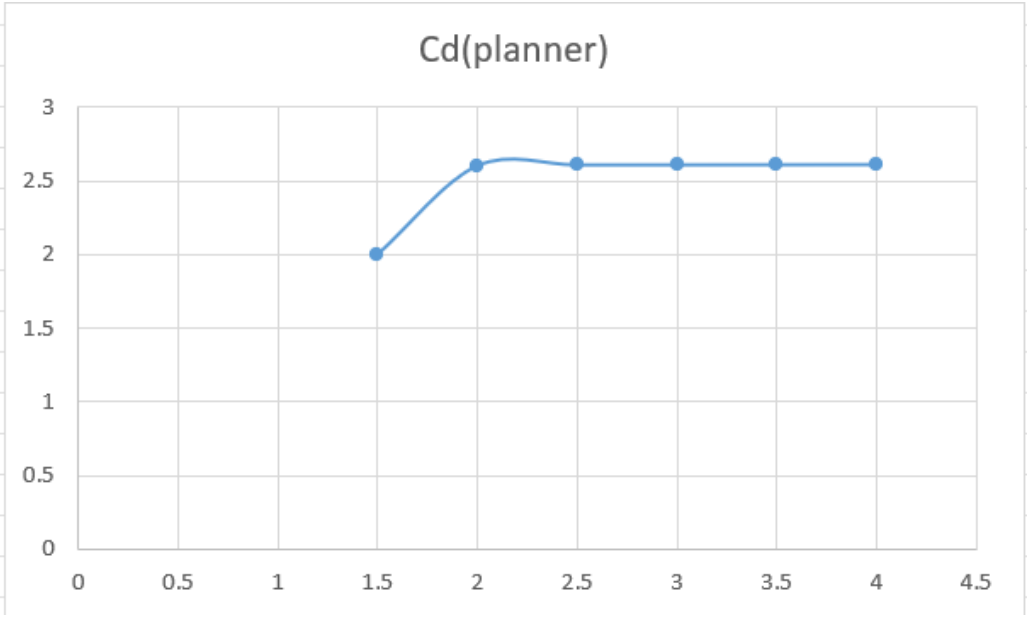
Results

For 4 bar total inlet pressure, pressure contour and velocity vector:





Co-efficient of discharge



Note: It is wrong as the Cd value cant be more than 1

Conclusion

- Cd value for the axisymmetric case is just about right but for the planner case it was not satisfactory. Better convergence for the planner case may have resolved the issue.

Appendix

- Cd values were calculated using a custom calculator, programmed in python

